

Description and Characteristics of Ocular Tumor Lysis Syndrome

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Keywords

Uveal melanoma · Ocular tumor lysis syndrome · Ocular oncology

Abstract

Introduction: The aim of the study was to describe and evaluate characteristics of ocular tumor lysis syndrome (OTLS) in eyes with uveal melanoma. **Methods:** Retrospective chart review of all patients with OTLS at the University of Colorado from 2009 to 2021. Data collected included patient demographics, tumor characteristics, radiation dosimetry, gene expression profiling (GEP), OTLS characteristics, management, and outcomes. **Results:** Seven eyes of seven patients with uveal melanoma treated with I-125 brachytherapy developed OTLS. Average age was 59 years (range 32–83). Mean apical height was 8.6 mm (range 6–11); mean diameter was 12.7 mm (range 8.5–15.3). All tumors were treated with plaques ≥ 16 mm in diameter. On presentation, 5/7 tumors had subretinal fluid, and 6/7 had collar-button configuration. OTLS presented as extensive pigment dispersion in the vitreous in all eyes, subretinal pigment and/or retinal detachment in 4/7 eyes, vitreous hemorrhage in 2/7 eyes, and anterior chamber pigment in 3/7 eyes. Four tumors were GEP class 1, two were class 2, and one was unclassified. Biopsy route was trans-scleral in 4/7 eyes and trans-vitreous in 3/7 eyes. OTLS occurred 2–4 weeks after an intraocular procedure in 5/7 eyes. All underwent pars plana vitrectomy. Cytology of the vitreous, obtained in five cases, showed pigment laden macrophages and hemorrhage, but

only 1/5 eyes had viable malignant cells. Four eyes were stable at the last follow-up, two were enucleated, and one had no light perception from pigmentary glaucoma. Poor vision ($<20/200$) occurred in 6/7 cases. Three patients died from metastasis (tumors were GEP class 2, GEP class without subclassification, and no GEP classification performed).

Conclusions: OTLS is a rare but devastating complication of uveal melanoma. Common characteristics included large plaque diameter, presence of subretinal fluid, and collar-button shape. The extensively dispersed pigment is typically not malignant. Though poor vision is common, enucleation may be avoided in most eyes through vitreoretinal surgical repair.

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Introduction

Uveal malignant melanoma is a disease with high morbidity and mortality, well studied in the Collaborative Ocular Melanoma Study (COMS), among others [1, 2]. Local tumor control methods are highly successful. However, complications from the various treatment modalities are common and metastasis occurs in approximately 35–50% of patients within 10 years regardless of treatment type [3, 4]. Survival after metastasis does not vary based on size of tumor and is only 20% at 1 year and 8% at 2 years following metastasis identification [5].

Presented at meetings: The Retina Society Annual Meeting 2021.

Brachytherapy offers globe-salvaging treatment in eyes with uveal melanomas. However, brachytherapy has many well-described potential complications including radiation retinopathy and optic neuropathy, undertreatment, regrowth, vitreous seeding, cataract, keratitis, rubeosis iridis, and neovascular glaucoma [6–9]. Other less understood complications include toxic tumor syndrome (ischemia resulting in exudative retinal detachment and neovascular glaucoma) and acute tumor-lysis related ocular hypertension (dispersion of golden-brown pigment and ocular hypertension after inappropriately high radiation doses) [10, 11]. We believe ocular tumor lysis syndrome (OTLS) is another distinct vision-threatening complication that is rarely characterized in the current literature, mainly in a few case reports that note necrosis of the uveal melanoma with overlying pigment dispersion and/or vitreous hemorrhage [12–14]. The purpose of this study was to describe OTLS in eyes with uveal melanoma and evaluate the predisposing factors, characteristics, and treatment of this condition.

Materials and Methods

This study was approved by the Institutional Review Board of the University of Colorado as an exempt retrospective review. The research adheres to the tenets of the Declaration of Helsinki and complies with the Health Insurance Portability and Accountability Act. This is a retrospective chart review of all patients with uveal melanoma who were diagnosed with OTLS between 2009 and 2021 by a single physician (S.C.N.O.). Data collected included patient age, sex, and affected eye.

Tumor characteristics included tumor location, tumor height and basal diameter, tumor description and presentation, presence or absence of collar button configuration, presence or absence of subretinal fluid (defined as at least one clock-hour of fluid), radiation dosimetry data, gene expression profiling (GEP), date of surgery, route of biopsy (trans-scleral vs. trans-vitreous), and associated surgical procedures including I-125 plaque placement, pars plana vitrectomy (PPV), and use of silicone oil. The date of OTLS diagnosis, presenting signs and symptoms, interim history between I-125 plaque therapy and OTLS diagnosis (additional intraocular surgeries/procedures and complications), time interval between I-125 plaque therapy and OTLS diagnosis, management of OTLS and vitreous cytology results, final visual acuity and anatomic results, need for enucleation, metastatic disease, and/or death were collected.

Results

Patient Characteristics

Seven eyes of 7 patients with uveal melanoma that were treated with Iodine-125 plaque brachytherapy and developed OTLS between January 2009 and August

2021 were included in this study. A total of 229 I-125 plaque brachytherapy procedures were performed over this same period. Of the OTLS cases, the average age was 59 years (range 32–83 years). Six patients were men, one was a woman. All 7 patients were Caucasian. The length of follow-up averaged 35.7 months (range 25–52 months) from the initial diagnosis to the last follow-up visit.

Tumor Characteristics

The mean apical height of the uveal melanoma was 8.6 mm (range 6–11 mm); the mean diameter was 13.1 mm (range 8.5–15.3 mm). The estimated spherical cap tumor volume averaged 180.5 mm³ (range 65.4–294.0 mm³). The center of the tumor was behind the equator in one eye, equatorial in three eyes, and far peripheral in three eyes. The tumor extended to the ciliary body in 5/7 eyes, and involved the macula in 4/7 eyes. One tumor was nasal, one was inferonasal, one was superonasal, one was inferior, one was inferotemporal, and two were temporal in location (Fig. 1). Six were classified as medium size and one was classified as large size based on COMS criteria [15–20]. All tumors were T2 or T3 using the American Joint Commission on Cancer version 8 (AJCC) classification [21]. On presentation, 5/7 (71%) had at least one clock-hour of subretinal fluid and 6/7 (86%) had a collar-button configuration. There was an average of 3.5 clock hours of exudation (range 0–7).

Radiation and Surgical Characteristics

All tumors were treated with radioactive plaques that were 16 mm or greater in diameter (two were 16 mm, one was 18 mm, four were 20 mm). The total radioactive energy to the eye averaged 82 mCi (range 38.0–113.5 mCi). Cumulative apical dose ranged from 82.7 to 96.0 Gy (average 89 Gy), and apical dose rates ranged from 51.4 to 83.0 cGy/h (average 71 cGy/h). The average dose to the macula was 41 Gy (range 18.9–69.0 Gy) and the average dose to the optic nerve was 33 Gy (range 25.0–49.9 Gy). The number of seeds used in the plaque averaged 20 (range 13–24 seeds). The biopsy route was trans-scleral in 4/7 eyes and trans-vitreous in 3/7 eyes. Four tumors were GEP class 1 (three were 1A, one was not subclassified as the test was performed prior to 1A/1B availability), two were class 2, one was unclassified. Three of the seven eyes underwent PPV with silicone oil instillation at the time of plaque placement to decrease the risk of radiation retinopathy due to the close proximity to the posterior pole. One additional patient underwent PPV at the time of plaque placement for a small amount of localized non-clearing vitreous hemorrhage overlying the tumor. It is significant to note that we take care to

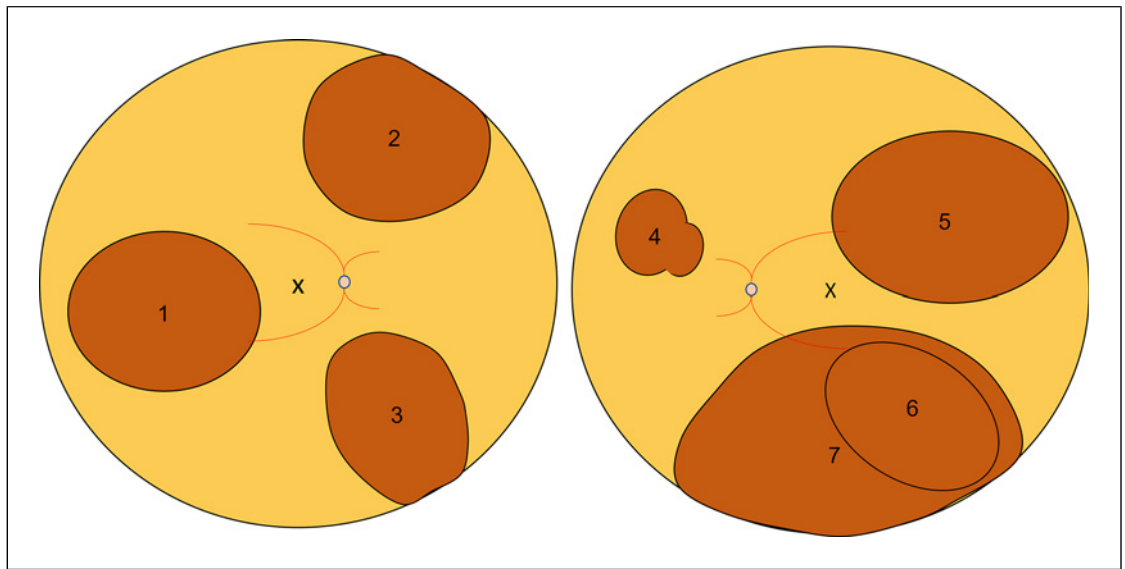


Fig. 1. Schematic showing a compilation of the approximate locations of all seven tumors within the retina.

avoid PPV port placement in the bed of the tumor itself when the tumors are anterior in location. Of the five anterior tumors in this series, only two underwent PPV (both at the time of plaque placement) prior to OTLS occurring and care was taken to avoid trocar insertions near the tumors.

OTLS Characteristics

OTLS occurred prior to surgical plaque placement in two eyes with ciliary body melanomas, 2–4 weeks after an intraocular procedure in five eyes (one of which also occurred prior to surgical plaque placement previously listed), and during tumor regrowth in one eye (2 years after plaque brachytherapy and after prior tumor regression). The five intraocular procedures preceding OTLS were one PPV/trans-vitreous biopsy/plaque placement/silicone oil/endolaser, one plaque placement with trans-scleral biopsy, one trans-scleral biopsy alone (prior to plaque placement 3 weeks later), and two cataract extractions with silicone oil removal (9 and 15 months after plaque brachytherapy). OTLS presented with extensive pigment dispersion in the vitreous in all eyes, subretinal pigment and/or retinal detachment in 4/7 eyes, vitreous hemorrhage in 2/7 eyes, and pigment in the anterior chamber in 3/7 eyes. Figures 2 and 3 provide two visual examples of cases of OTLS, depicting images of the eyes before OTLS occurred and then images of how OTLS presented. All eyes had intraocular pressure within the normal range on presentation. PPV was performed in all cases for visual rehabilitation and diagnostic purposes.

The main goal of PPV was to remove pigment and to repair any associated retinal detachment. A second PPV was required in 3/7 eyes for management of recurrent rhegmatogenous retinal detachment not directly related to OTLS but from retinal breaks and proliferative vitreoretinopathy. During the PPV, the tumor was managed in two cases with partial endoresection of the necrotic portions of the tumor due to noting significant intraoperative friability. In four cases, long-duration endolaser was applied to the tumor, including one of the eyes that also underwent endoresection. All seven eyes achieved anatomic success at final visit (or prior to enucleation), defined as retinal attachment without recurrent evidence of OTLS. Cytology of the vitreous, which was obtained in five cases during PPV, typically showed pigment laden macrophages and hemorrhage, but only 1/5 had viable malignant cells. An additional case was not definitive with rare mildly atypical small cells with enlarged nuclei. This eye was later enucleated for intractable pain and pathology demonstrated no residual melanoma. Four of seven eyes were stable at the last follow-up, two were enucleated, and one had no light perception due to secondary pigmentary glaucoma. Poor vision (<20/200) occurred in 6/7 cases, including the two eyes that had poor vision prior to enucleation.

Three patients were later diagnosed with metastatic disease, one of which was GEP class 1 (assessed prior to availability of 1A/1B designation), one was GEP class 2, and one was not tested per patient request. These patients passed away from metastatic uveal melanoma.

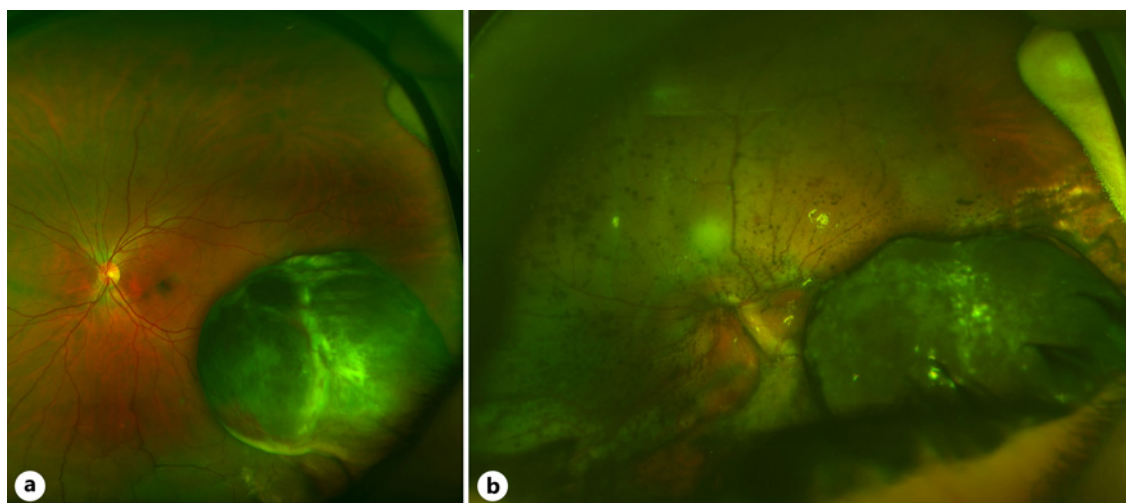


Fig. 2. Wide-field color fundus photographs of an eye with choroidal melanoma before treatment with plaque brachytherapy with PPV, trans-vitreous biopsy, silicone oil insertion, endolaser, cryotherapy, and prior to ocular tumor lysis (a) and 15.5 months later when ocular tumor lysis occurred (b). Prior to plaque brachy-

therapy and tumor lysis, the tumor exhibited a collar-button configuration and adjacent inferior subretinal fluid. Ocular tumor lysis presented with extensive subretinal and preretinal pigment dispersion, tumor necrosis with overlying retinal necrosis, and a total exudative retinal detachment.

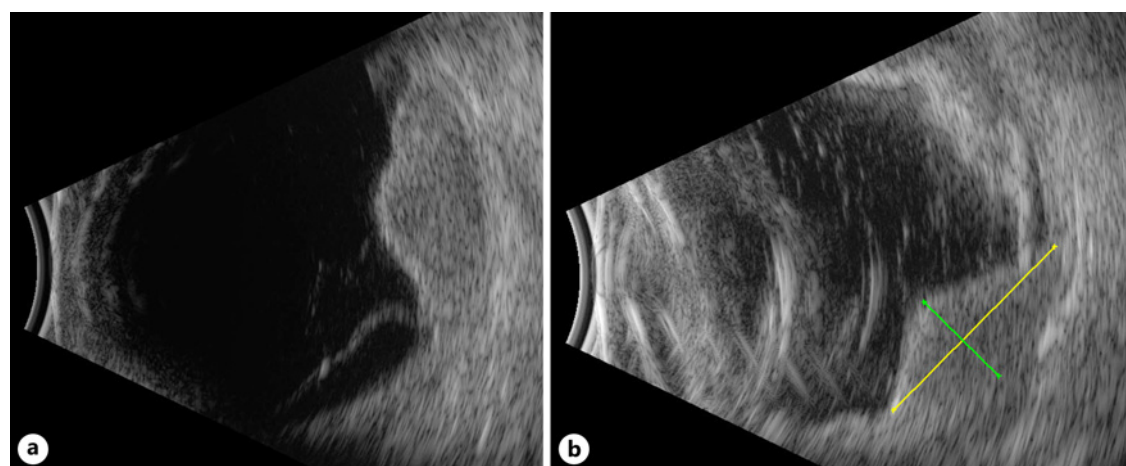


Fig. 3. B-scan ultrasonography of an eye with choroidal melanoma before treatment with plaque brachytherapy with PPV, trans-vitreous biopsy, silicone oil insertion, and prior to ocular tumor lysis (a, L2) and 13 months later when ocular tumor lysis occurred (b, T2PE). Prior to plaque brachytherapy and tumor lysis, the tumor exhibited a collar-button configuration and adjacent inferior subretinal fluid. Ocular tumor lysis presented as a fi-

brinous membrane on the anterior lens surface and pigment dispersed on the posterior lens surface, posterior synechiae, extensive pigment in the vitreous, pigment dispersed throughout the surface of the retina and subretinal space, pre-retinal cellular membranes, tumor necrosis, and an inferonasal rhegmatogenous retinal detachment from an atrophic retinal hole overlying the tumor.

Discussion

In this retrospective review, we evaluated the characteristics of eyes with OTLS. Potential OTLS risk factors included recent intraocular surgery, large tumor and plaque diameter, presence of subretinal fluid, and tumor

collar button configuration. This information suggests that OTLS may be more common in eyes with highly active and larger tumors. Many of the tumors were broad, extending to the ora and/or into the macula.

It is interesting to note that OTLS occurred 2–4 weeks after plaque treatment in two eyes, and 2–4 weeks after an

additional intraocular procedure in three eyes. This raises the question whether intraocular surgery increases the risk of OTLS and therefore if surgical procedures should be used judiciously in eyes that have other risk factors for OTLS. It is important to understand possible risk factors for OTLS in order to better mitigate the risk of developing this complication when possible, and to provide patients with more accurate risk and prognostic information.

The current literature on OTLS includes a few case reports with similar findings to those seen in this series, but to our knowledge, there is no large descriptive series. We believe OTLS as presently described is distinct from previously described entities dubbed acute tumor-lysis related ocular hypertension and toxic tumor syndrome. A series describing acute tumor-lysis related ocular hypertension has many key differences from our series [11]. Tumor lysis in their series was correlated with an inappropriate overdose of radiation. In their series, tumor lysis occurred with high apical dose rates of 118–289 cGy/h, while the complication did not occur in eyes irradiated with dose rates under 109 cGy/h. In contrast, our OTLS patients underwent lower apical dose rates of 51.4–83.0 cGy/h. Their series also noted that tumor lysis was a relatively late complication, occurring after a mean of 17.4 months post plaque treatment. This was different from our bimodal distribution of within 4 weeks and 12–19 months after plaque treatment. The eyes in their series also all presented with secondary ocular hypertension. This is in stark contrast to our series in which all the eyes had normal presenting intraocular pressure, but two later developed a secondary pigmentary glaucoma. Finally, their series described OTLS as dispersion of golden-brown or rust-colored cellular debris, but our patients had primarily dark pigment dispersion. It is possible that tumor-lysis related ocular hypertension is a variant of OTLS associated with very high doses of radiation.

Another rare previously described complication of irradiated uveal melanomas, toxic tumor syndrome, has been noted previously as a mainly ischemic process thought to involve the release of inflammatory mediators. This disease has been characterized by the presence of exudative retinal detachment and neovascular glaucoma [10, 22]. In contrast to our series, toxic tumor syndrome has not been associated with significant pigment dispersion, a major defining characteristic of OTLS. Further, while OTLS can have later secondary pigmentary glaucoma, none of our patients presented with ocular hypertension or neovascular glaucoma, as seen in toxic tumor syndrome. While toxic tumor syndrome is typically treated with tumor endoresection, our eyes with

OTLS could be managed and stabilized without complete endoresection. All of our eyes with OTLS achieved anatomic success at final visit (or prior to enucleation) despite only partial endoresection in two cases, endolaser to the tumor in four cases (including one that also underwent partial endoresection), and no specific tumor management in two eyes. The main goal of PPV was to remove pigment from the visual axis and overlying the tumor and repair any associated retinal detachment to maximize visual recovery and enable adequate tumor monitoring. Overall, our current series and the prior descriptions of toxic tumor syndrome differ in presentation and management, suggesting that OTLS is a distinct complication from toxic tumor syndrome.

While many of the eyes in this case series had poor visual outcomes, many were spared enucleation due to adequately differentiating OTLS from active tumor or regrowth. Vitreous specimens can differentiate melanophages from viable tumor cells, which look similar on clinical examination [23]. In our series, only 20% of the eyes that underwent cytology had viable malignant cells in the vitreous. In previous studies, viable tumor cells have been found in the vitreous of eyes with adequately treated tumors about 13–14% of the time, which corresponds approximately with our findings here [24, 25]. Further, viable malignant cells and mitotic activity are found in most if not all stable irradiated tumors, albeit in lower numbers than in nonirradiated tumors or irradiated tumors with evidence of regrowth [26–29]. We recommend vitrectomy with vitreous sample biopsy for therapeutic and diagnostic purposes for eyes with suspected OTLS. Vitreoretinal surgical repair is the definitive treatment for OTLS to clear the vitreous opacities and repair any associated retinal detachment to achieve global salvage.

Two eyes in our series were enucleated for reasons unrelated to OTLS. One eye was enucleated due to light perception vision with low visual potential and suspected tumor regrowth 1.5 years after OTLS. No vitreous biopsy had been performed in this case, so it is unknown if there were viable malignant cells at the time of OTLS. The tumor had shown initial signs of regression and shrinkage after OTLS but developed a new lesion inferiorly in the anterior chamber angle that was releasing pigment thought to be an extension of the ciliary body melanoma with possible seeding 1.5 years after OTLS. Pathology at the time of enucleation demonstrated metastatic deposits throughout the vitreous cavity. The second eye that was enucleated had intractable pain with hand motion vision and low visual potential. The vitreous biopsy performed for OTLS management had shown rare mildly atypical small cells with enlarged nuclei and abundant pigmented

melanophages, but pathology at the time of enucleation showed no evidence of residual melanoma.

Importantly, there was a high rate of death from metastatic disease in this study compared with prior rates noted in patients with uveal melanoma. Three of the 7 patients (43%) passed away during the follow-up period – one of which was GEP class 1 (assessed prior to availability of 1a/1b designation), one was GEP class 2, and one was not tested per patient request. The follow-up period was an average of 3 years from the initial diagnosis to the last visit to our center, or an average of 6 years from the initial diagnosis to date of data collection. Prior studies have shown an 18–32% 5-year mortality rate of patients with irradiated medium-sized uveal melanomas, but that larger tumor diameter is correlated with lower survival rates [5, 17, 30]. Given the higher mortality rate in this study, we query if patients with OTLS are at higher risk for metastatic disease related to this complication or if patients with OTLS tend to have larger tumor diameters that are more likely to metastasize independent of OTLS. Given the low rate of viable tumor cells in the vitreous and evidence of tumor shrinkage in all our cases, we hypothesize that the high mortality rate is more likely due to the tumors' size. Larger studies would be needed to better answer this important question.

OTLS is a rare but devastating complication of uveal melanoma. In these OTLS cases, common characteristics included larger tumor size, presence of subretinal fluid, and collar-button shape. Understanding this complication may help mitigate risk and understand the poor visual and systemic prognosis of patients with OTLS. The extensively dispersed pigment cells are typically not malignant. Therefore, while poor vision is common, enucleation may be avoided in most eyes with OTLS through vitreoretinal surgical repair of hemorrhage, pigment, and retinal detachment.

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Statement of Ethics

This study was approved by the Institutional Review Board of The University of Colorado as an exempt retrospective review. The research adheres to the tenets of the Declaration of Helsinki and complies with the Health Insurance Portability and Accountability Act. Written consent was not required because this study was approved by the Institutional Review Board of The University of Colorado as an exempt retrospective review. The research adheres to the tenets of the Declaration of Helsinki and complies with the Health Insurance Portability and Accountability Act.

Conflict of Interest Statement

There are no financial conflicts of interest for any of the authors.

Funding Sources

The study received Challenge Grant from Research to Prevent Blindness (RPB) and from the Department of Ophthalmology, University of Colorado.

Author Contributions

Talisa E. de Carlo Forest contributed to data collection, data analysis, and manuscript drafting. Scott C.N. Oliver contributed to data analysis and manuscript editing.

Data Availability Statement

All data generated or analyzed during this study are included in this article. Further enquiries can be directed to the corresponding author.

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